

7	Electrons are emitted from a metal surface when it is illuminated with electromagnetic radiation of suitable frequency. This process is known as the photoelectric effect.								
	(a)	State two observations of the photoelectric effect that cannot be explained using the wave nature of electromagnetic radiation. 	[2]						
	(a)	Solution: The photoelectron emission is almost instantaneous even if light of very low intensity is used. At constant frequency of electromagnetic radiation, the maximum kinetic energy of the photoelectrons is independent of the intensity of the radiation. No photoelectron is emitted if the frequency of the electromagnetic radiation used is lower than a certain threshold value, even if light of high intensity is used.	1 1						
	(b)	Data for the wavelength λ of the radiation incident on the metal surface and the maximum kinetic energy E_K of the emitted electrons are shown in Fig. 7.1. <table><tr><td>λ / nm</td><td>E_K / 10^{-19} J</td></tr><tr><td>650</td><td>-</td></tr><tr><td>240</td><td>4.44</td></tr></table> Fig. 7.1	λ / nm	E_K / 10^{-19} J	650	-	240	4.44	
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	(i)	Suggest why emitted electrons are likely to have a range of values of kinetic energy for any one frequency of the electromagnetic radiation. 	[2]						
	(i)	Solution: An electron at the surface of a metal requires the least amount of energy to be liberated. Thus, it escapes with the maximum possible kinetic energy. Electrons below the surface emerge with less kinetic energy. As these electrons are on their way out to the surface, they lose part of the kinetic energy gained from the radiation due to collision and work done against the electrostatic forces of attraction.	1 1						
	(ii)	Without any calculation, suggest why no value is given for E_K for radiation of wavelength 650 nm. 							

8	Read the following article and answer the questions that follow	
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